

CSCC63 – WEEK 6 TUTORIALS

1. Do question 7.9 from Sipser:

A **triangle** in an undirected graph is a 3-clique. Show that $\text{TRIANGLE} \in \mathcal{P}$, where

$$\text{TRIANGLE} = \{ \langle G \rangle \mid G \text{ contains a triangle} \}$$

See pages 295-296 for the definition of cliques.

2. Do question 7.13 from Sipser:

Let $\text{MODEXP} = \{ \langle a, b, c, p \rangle \mid a, b, c, \text{ and } p \text{ are positive binary integers, and } a^b = c \pmod{p} \}$

- Why is the obvious algorithm to solve this problem not polytime?
- Show that this language is in $\mathcal{P}$.

3. Show that $\mathcal{P}$ is closed under:

- union,
- complement,
- concatenation, and
- (Extra!) Kleene Star.

If you need to review the Kleene star operation, the Kleene star L^ of a language L is defined as follows. For any language L :*

- $L^0 = \{\varepsilon\}$.
- For any $i \in \mathbb{N}$, $L^{i+1} = \{xy \mid (x \in L^i) \wedge (y \in L)\}$.
- $L^* = \bigcup_{n=0}^{\infty} L^n$.